

Gun Kaewngarm

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EXPERIENCE

Senior Software Engineer

Jan 2024 - Ongoing

Adrich, LLC (In partnership with Carnegie Mellon University) | Pittsburgh, PA, USA

- Led the architecture redesign of Adrich's backend to scale IoT Smart Labels from 20,000 to 5 million devices using Node.js (Express) and Python (FastAPI).
- Developed a data transfer protocol for smart labels by implementing ECL compression in C on devices, caching, and decompression via Node.js on the backend.
- Designed and implemented a calibration dashboard using React to streamline the setup of IoT Smart Label devices.
- Designed and implemented CI/CD pipelines using GitLab CI and deployed solutions on Digital Ocean.

Software Engineer

June 2021 – June 2023

Associate Software Engineer

June 2020– May 2021

Agoda Services Co., Ltd., | Bangkok, Thailand

- Led the project to containerize services using Docker, Shell script, and Helm for deployment in the company's experimental Kubernetes clusters to minimize developer testing and deployment time.
- Designed and developed a campaign engine service to create, schedule, and execute marketing campaigns to target multiple audiences by querying data from various sources such as SQL and Hadoop. Also, help the business team develop queries to fetch audiences from those data sources.
- Designed and developed data logging and tracking processes to analyze customer behaviors. This data is used to create more relevant marketing campaigns tailored to meet booker behaviors.
- Developed a service using Spark to fetch traffic from the website and identify which user to send reminder retargeting emails to promote booking on the platform. This results in more revenue for the company.
- Implemented and maintain a Continuous Integration pipeline using Gitlab CI for all projects.

Software Developer Intern

June 2019 – July 2019

Demeter ICT | Bangkok, Thailand

- Developed an embedded Javascript application on the Zendesk platform for customer service agents.
- Developed features of prototype Android applications using Java and tuning its existing features.

EDUCATION

Carnegie Mellon University | PA, United States

Master of Software Engineering

Expected Dec 2024

- GPA: 4.05
- Course: Cloud Computing, DevOps & Security, Web Applications, Software Architectures, Requirements for Information System, Quality Assurance, Agile Methods, Project Management

Chulalongkorn University | Bangkok, Thailand

Bachelor of Engineering (Computer Engineering)

June 2020

- GPA: 3.65
- Course: Introduction to data structures, Algorithm Design, System analysis and design, Software engineering, Cloud Computing, Data science and data engineering, Computer Security

HONORS AND AWARDS

Bank of Thailand Scholarship | Bank of Thailand

September 2022

- A full-ride scholarship granted by the Bank of Thailand to study for a master's degree focusing on software engineering abroad.

SKILLS

- **Languages:** Scala, Java, Python, JavaScript, SQL (T-SQL, MYSQL, SparkSQL)
- **Frameworks:** Node.js, Fast API, Flask, Akka, Spring
- **Cloud platform:** Amazon Web Services (AWS), Google Cloud Platform (GCP), Microsoft Azure
- **Tools/Others:** Gitlab CI, Jenkins, Docker, Helm, Kubernetes, Git, Linux, Teamcity, Consul, Vault, Shell Scripting, Impala, Hive, Hadoop, Spark, Kafka

PROJECTS

Real-time Multiplayer Battleship Game | Associated with Carnegie Mellon University

- Developed a real-time multiplayer game with a JavaScript and HTML front-end and Python (Django) backend using WebSocket for real-time communication.
- Implemented features for user registration, login, matchmaking, and in-game chat.
- Enabled real-time gameplay with support for multiple concurrent matches.
- Deployed on Amazon EC2 with SSL certification for secure connections and scalability.

Rama IV Traffic Dashboard System ([link](#)) | Associated with Chulalongkorn University

- Developed a traffic monitoring dashboard and traffic speed prediction system using taxis' GPS data.
- Implemented the dashboard using Plotly Dash with Google BigQuery for big data storage and deployed on Google Cloud Run for containerized scalability.
- Created a traffic speed prediction system using a CNN-based model that treats traffic data as images, translating spatiotemporal dynamics into time-space matrices for accurate predictions. Achieved a speed deviation of approximately 10 km/h.
- Deployed the prediction model using Flask, packaged in a Docker container, and deployed to the cloud using Google Cloud Run, with data stored in Google BigQuery.

Namo Chat Bot ([link](#)) | Associated with Chulalongkorn University

- Developed a Dhamma chatbot in the LINE application to assist individuals with depression and promote mindfulness practices.
- Utilized Google Cloud products including Dialogflow for natural language understanding, Cloud Scheduler for task automation, Cloud Datastore for data storage, and Cloud Storage for file management.
- Containerized the application with Docker and deployed it on Google Cloud Run for scalable performance.